Pricing AT1 Contingent Convertibles as a Claim on Bank Capital

Model, numerical method, and a calibration to BNP Paribas's USD AT1s

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revised October 2026: Tier 2 anchor, MDA path, robustness, data layout,
Tier 2 restriction on the capital dynamics

Abstract

An Additional Tier 1 (AT1) bond is a perpetual, deeply subordinated bank bond whose cash flows depend on the issuer's capital: coupons are cut when capital enters the regulatory buffer, principal is converted or written down at a capital trigger or at the regulator's discretion, and the issuer calls only when refinancing is cheaper. We take the Common Equity Tier 1 (CET1) ratio as the state variable, add a mean-reverting issuer spread that drives both the call decision and the regulatory write-down (PONV) hazard, and price by Monte Carlo with the PONV hazard integrated analytically along each path. Calibrated to BNP Paribas's three USD AT1s on 25 September 2026, with the FRED Treasury curve, 50 quarters of CET1 history and an MDA threshold that rises with the scheduled O-SII step-ups, the three bonds imply a hazard of about 2.2% a year, equal across bonds to within the bid–offer. Pricing BNP's Tier 2 and senior non-preferred bonds on the same simulated state splits that hazard: 1.7% is priced in Tier 2 as well (resolution risk plus Tier 2's own risk and liquidity premia, at a 20% recovery). Of the $\approx$ 270–280bp spread, about 180bp is priced in Tier 2, a figure robust to a reputational call rule, to the order of the decomposition, to a term structure of the resolution hazard and to the dynamics of capital. The remaining 89–98bp, equally robust, pays for being an AT1 rather than a Tier 2: expected MDA coupon cuts, conversion, extension and an AT1-specific premium. Its split is not robust. The speed of capital mean reversion is the weakest input (the half-life of a capital shock is anywhere between 1.5 and 11 years), and the AT1-specific premium is about 50bp at our base case but zero at the historical estimate, where coupon cuts and extension take its place. The Tier 2 restricts it: pricing the AT1 as safer than the Tier 2 would violate the creditor hierarchy, which rules out the slower half of the plausible range. The price is flat in capital above the MDA threshold ($\approx$ 0.6 points per point of CET1) and steep at it ($\approx$ 4 points). The data end on 25 September 2026, before BNP's spreads widened with French sovereign stress in October.

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1 The instrument and the idea

1.1 What an AT1 is

An AT1 bond counts as going-concern regulatory capital for a bank. Its terms make it behave partly like equity:

1. **Perpetual, with issuer calls.** There is no maturity. The issuer may redeem at par on call dates; for the bonds studied here, only on *reset dates*, every five years from the first call. If not called, the coupon resets to a reference rate (5-year US Treasury, CMT) plus the

margin fixed at issue. Investors are exposed to *extension*: the issuer calls when it is cheap to refinance, and does not when it is expensive, which is exactly when investors would want their money back.

2. **Discretionary, non-cumulative coupons.** Coupons may be cancelled at any time and are lost for good. They are also mechanically restricted when CET1 falls inside the *combined buffer requirement*: the Maximum Distributable Amount (MDA) rule of CRD IV.

3. **Loss absorption.** If CET1 falls below a contractual trigger (5.125% here), the bond converts into shares (or is written down). The resolution authority can also impose losses at the *Point of Non-Viability* (PONV) while CET1 is still above the trigger: Credit Suisse's AT1s were written to zero in March 2023 with a reported CET1 ratio of about 14%.

1.2 Why capital is the state variable

Trigger, coupon restriction and (partly) the call decision are all written on the CET1 ratio. Modelling CET1 directly makes these three risks consistent by construction: the same simulated path that approaches the MDA threshold also cuts coupons, discourages a call, and raises the chance of loss absorption. The alternative, an equity-barrier model, needs a mapping from share price to capital that is unobservable and unstable.

Capital alone is not enough, however. The market prices AT1s far below what historical capital volatility would justify (§7), and the call decision depends on market spreads, not only on capital. We therefore add a second state variable, the issuer's AT1 spread, which (i) drives the call decision and (ii) scales the PONV hazard.

The PONV hazard then absorbs everything the mechanical features leave unexplained, and on its own it cannot say how much of that is the bank's credit risk and how much is specific to the AT1. The bank's other bonds can. Under EU resolution the creditor hierarchy holds: CET1 absorbs losses first, then AT1, then Tier 2. BNP's Tier 2 bonds are exposed to the same resolution event as its AT1s, without coupon discretion, MDA restrictions, trigger or perpetual extension. Pricing them on the same simulated state (§3.5, §7.3) splits the AT1 hazard into a part already priced in Tier 2 and an AT1-specific premium.

2 State dynamics

2.1 CET1 ratio

Let C_t be the CET1 ratio in percentage points. We use a mean-reverting jump-diffusion with downward jumps:

$$dC_t = \kappa(\theta - C_t)\,dt + \sigma\,dW_t - J\,dN_t, \tag{1}$$

where N_t is a Poisson process with intensity λ_J (stress events) and jump sizes are i.i.d.

$$J = m_J \exp\!\big(\sigma_J Z - \tfrac{1}{2}\sigma_J^2\big), \qquad Z \sim N(0,1), \qquad \mathbb{E}[J] = m_J.$$

Mean reversion to θ represents the bank's capital management (retained earnings, distribution policy, management of risk-weighted assets); the target θ is management's stated CET1 objective. Jumps are one-sided because losses arrive suddenly while recapitalisation is gradual.

The *mechanical trigger time* is $\tau_m = \inf\{t :\ C_t \le c_{\text{trig}}\}$.

2.2 Issuer AT1 spread

Let S_t be the market level of the issuer's new-issue AT1 spread (decimal). It follows an Ornstein–Uhlenbeck process correlated with capital shocks,

$$dS_t = a(\bar{s} - S_t)\,dt + \eta\,dB_t, \qquad d\langle B, W\rangle_t = \rho\,dt, \quad \rho < 0, \tag{2}$$

and the issuer's *effective* spread adds a capital premium once CET1 is inside the buffer:

$$s_t = S_t + \beta\left(\bar{c}_{\mathrm{MDA}} - C_t\right)^+, \tag{3}$$

where $\bar{c}_{\mathrm{MDA}}$ is the MDA threshold defined below. The effective spread s_t is the rate at which the bank could issue a new AT1 at time t.

3 Contract features as functionals of the state

3.1 MDA coupon restriction

Let b be the MDA floor (Pillar 1 plus the CET1 part of Pillar 2R) and w the combined buffer (conservation, systemic and countercyclical buffers), so that $\bar{c}_{\mathrm{MDA}} = b + w$. With $q = (C - b)/w$ the position inside the buffer, the fraction of the scheduled coupon that can be paid is

$$\phi(C) = \begin{cases} 1 & C \geq b + w, \\ 0.6 & q \in [0.75, 1), \\ 0.4 & q \in [0.50, 0.75), \\ 0.2 & q \in [0.25, 0.50), \\ 0 & q < 0.25 \text{ or } C < b. \end{cases} \tag{4}$$

This is a stylised version of the quartile schedule; it makes coupon-cancellation risk emerge from the same paths as trigger risk.

3.2 Call and extension

Let $T_1 < T_2 < \dots$ be the reset (call) dates and m the reset margin over the 5-year reference rate. The issuer calls at the first reset date T_k at which

$$m > s_{T_k} \qquad \text{and} \qquad C_{T_k} \geq \bar{c}_{\mathrm{MDA}} \qquad \text{and} \qquad T_k < \tau_m, \tag{5}$$

that is, when keeping the bond costs more than issuing a new one, provided the regulator would approve the call (CET1 above the MDA threshold) and the bond is still alive. Denote the call time by τ_c ($+\infty$ if never). If $\eta > 0$ the call is random even for a healthy bank; the low-margin bond is the first to be extended when spreads widen.

3.3 Conversion

At τ_m the bond converts into shares at the higher of the market price and a floor price P_{floor}. The investor receives shares worth $F \cdot S^{\mathrm{eq}}_{\tau_m}/P_{\mathrm{floor}}$ (when the share trades below the floor, as it will at the trigger). The model uses a stressed share price $S^{\mathrm{eq}}_{\tau_m} = \xi\, S^{\mathrm{eq}}_0 X_0$ (converted to USD at X_0), so the recovery is

$$R = \min\left(1,\ \frac{\xi\, S^{\mathrm{eq}}_0 X_0}{P_{\mathrm{floor}}}\right). \tag{6}$$

3.4 PONV: a hazard linked to the spread

The regulator may impose losses before the trigger. We model this as a doubly-stochastic (Cox) event with intensity proportional to the effective spread,

$$\lambda_t = \lambda \, \frac{s_t^+}{s_0}, \tag{7}$$

so λ is today's hazard (the Gaussian spread is floored at zero, so the hazard is never negative), and the hazard rises when spreads widen or capital falls inside the buffer. Recovery at PONV is zero. Linking the hazard to s_t is what makes extension consistent: the states in which the issuer extends ($s_t > m$) are exactly those in which the bond is riskier and worth less. With a constant hazard, extension would be valued at a discount rate that does not reflect the state, and could even look valuable to the investor.

3.5 Tier 2 and senior bonds: the resolution hazard

Write the AT1 hazard as

$$\lambda = \lambda_{\text{res}} + \lambda_{\text{spec}}, \tag{8}$$

where λ_{res} is the hazard of BNP's resolution (the event that also hits Tier 2) and λ_{spec} is everything the AT1 pays for beyond it: a write-down while the bank is not resolved, the risk premium of a perpetual, discretionary, equity-like claim, and its liquidity. Both scale with the same driver s_t^+/s_0.

BNP's Tier 2 and senior non-preferred (SNP) notes are callable once, one year before maturity (or at a reset date), when they stop counting as capital or MREL; the market prices that call as certain, so each is a bullet to its call date T with coupons c_i. At resolution the holder receives a fraction R of face. With the Tier 2 exposed to λ_{res} only,

$$B_0 = \mathbb{E}\left[\sum_{t_i \le T} c_i \, e^{-r t_i - \lambda_{\text{res}} I_{t_i}} + F e^{-rT - \lambda_{\text{res}} I_T} + RF \sum_{t_n < T} \left(e^{-\lambda_{\text{res}} I_{t_n}} - e^{-\lambda_{\text{res}} I_{t_{n+1}}}\right) e^{-r t_{n+1}}\right], \tag{9}$$

where the last sum is the conditional probability of resolution in each grid step times the discounted recovery. Like (11), it is analytic in the hazard on a fixed set of paths, so λ_{res} is the root of $B_0 = P^{\text{clean}} + \text{AI}$, solved on the same paths and random numbers as the AT1s.

4 Valuation

4.1 Pricing equation

Let r be the flat, continuously compounded risk-free rate, F the face and t_i the coupon dates, paid at frequency f. The scheduled coupon is $c(t_i) = c_0$ up to and including T_1 and $c(t_i) = y^{\text{ref}} + m$ after it. Let τ_P be the PONV time. The dirty price is

$$\begin{aligned} V_0 = \mathbb{E}\Bigg[& \sum_{t_i} \mathbf{1}_{\{t_i < \tau_m,\ t_i \le \tau_c,\ t_i < \tau_P\}} \frac{\phi(C_{t_i})\, c(t_i)\, F}{f} \, e^{-r t_i} + \mathbf{1}_{\{\tau_c < \tau_m,\ \tau_c < \tau_P\}} \, F \, e^{-r\tau_c} \\ & + \mathbf{1}_{\{\tau_m < \tau_c,\ \tau_m < \tau_P\}} \, R F \, e^{-r\tau_m} \Bigg] + \text{tail}. \end{aligned} \tag{10}$$

Paths alive at the horizon $H = 40$ years receive the value of a perpetuity at the reset coupon, $F(y^{\text{ref}} + m)/(r + \lambda_H)$, discounted to today.

4.2 Conditional Monte Carlo for the PONV hazard

Proposition 1. *Let $\mathcal{G}$ be the σ-algebra generated by the paths of (C, S). Conditional on $\mathcal{G}$, the PONV time has intensity (7), so*

$$\Pr(\tau_P > t \mid \mathcal{G}) = \exp\Big(-\lambda \int_0^t \frac{s_u}{s_0}\, du\Big) \equiv e^{-\lambda I_t}.$$

Hence

$$V_0(\lambda) = \mathbb{E}\left[\sum_{t_i} \mathrm{CF}_i\, e^{-rt_i - \lambda I_{t_i}} + \mathrm{CF}_{\mathrm{end}}\, e^{-rT_{\mathrm{end}} - \lambda I_{T_{\mathrm{end}}}}\right], \tag{11}$$

where CF_i are the coupon cash flows and $\mathrm{CF}_{\mathrm{end}}$ is the call, conversion or tail payment at the end of each path's life, all computed without reference to τ_P.

Proof. Condition each indicator $\mathbf{1}_{\{t<\tau_P\}}$ in (10) on $\mathcal{G}$ (tower property); every other term is $\mathcal{G}$-measurable. □

Consequences for the numerics:

- PONV is never simulated as an event, which removes its sampling noise (Rao–Blackwellisation).
- On a fixed set of paths, $V_0(\lambda)$ is smooth and strictly decreasing in λ (because the driver starts at $s_0/s_0 = 1$, so $I_t > 0$ for every $t > 0$ on every path). The market-implied hazard is therefore the unique root of $V_0(\lambda) = P^{\mathrm{clean}} + \mathrm{AI}$ and is found with Brent's method.
- Cash flows and I_t are stored per path once; each evaluation of $V_0(\lambda)$ is a vectorised sum.

4.3 Discretisation

The grid is monthly ($\Delta = 1/12$) over 40 years, with 20,000 paths. CET1 uses an Euler step with a Bernoulli jump of probability $1 - e^{-\lambda_J \Delta}$; the spread uses the exact OU transition

$$S_{n+1} = \bar{s} + (S_n - \bar{s})e^{-a\Delta} + \eta\sqrt{\tfrac{1-e^{-2a\Delta}}{2a}}\,\big(\rho Z_n^C + \sqrt{1-\rho^2}\, Z_n^{\perp}\big).$$

Coupons are discounted at their exact dates; path events (trigger, call) are read on the grid. Market prices are clean, so the model value is compared with the clean price plus 30/360 accrued interest (the accrued amount matches Capital IQ to three decimals).

4.4 Common random numbers

All primitive random numbers are drawn once and reused across scenarios, bumps and feature switches. Differences between two prices (Greeks, stress P&L, spread components) therefore carry no Monte Carlo noise of their own.

5 Calibration

5.1 Capital dynamics from history

The CET1 series is quarterly from 2014Q1 to 2026Q2 (50 observations). Earlier data follow Basel II/2.5 definitions and are not comparable. The 2023Q1 change (+1.2pp, sale of Bank of

the West) is a disposal, not a capital shock, and that transition is excluded from the likelihood. The OU transition density is Gaussian for any step Δ_k,

$$C_{k+1} \mid C_k \sim N\Big(\theta + (C_k - \theta)e^{-\kappa\Delta_k},\ \tfrac{\sigma^2}{2\kappa}\big(1 - e^{-2\kappa\Delta_k}\big)\Big),$$

so the likelihood is exact. Fixing $\theta = 13\%$ (BNP's target) gives $\hat{\kappa} = 0.165$ and $\hat{\sigma} = 0.44\text{pp}/\sqrt{\text{yr}}$; κ is weakly identified and the base case uses $\kappa = 0.35$ (§7.8), with $\sigma = 0.47$ re-estimated given κ. The 2014Q2 fall of 0.76pp (US sanctions settlement) is an example of the stress events the jumps represent. The jumps cannot be identified from this sample; we use $\lambda_J = 0.12$, $m_J = 2.2$pp, $\sigma_J = 0.5$, whose 95th-percentile jump (4.4pp) matches BNP's CET1 depletion in the EBA 2025 adverse scenario (4.2pp, fully loaded).

5.2 Regulatory geometry

The floor is $b = 4.5 + 1.14 = 5.64\%$ (Pillar 1 and the CET1 part of Pillar 2R, 2025 stack). The combined buffer is set so that $b + w$ equals the latest published CET1 requirement, 10.43% at 30 June 2026 ($w = 4.79\%$: conservation, G-SIB/O-SII, countercyclical and systemic-risk buffers). The threshold is not constant: BNP's O-SII buffer rises by 25bp on 1 January 2027 and again on 1 January 2028, and the higher of the G-SIB and O-SII buffers applies, so w and $\bar{c}_{\text{MDA}}$ step up to 10.68% and 10.93% along every path. The trigger is $c_{\text{trig}} = 5.125\%$ of group CET1.

5.3 Market inputs

The inputs are:

- the clean price of each bond: the three AT1s, two Tier 2 notes (5.906% 2035NC34, 2.588% 2035NC30) and one SNP note (5.738% 2035NC34), all in USD (Capital IQ evaluated prices, bid–offer 0.07–0.29 points);
- the FRED constant-maturity Treasury curve (DGS5, DGS7, DGS10): the 5-year CMT is the reset reference, and each bond is discounted at the yield interpolated at its first call date;
- the AT1 spread S_0, the average of (bond yield − call-matched Treasury);
- the latest published CET1 ratio, with a 35-day publication lag;
- the share price;
- BNP's senior CDS (EUR, MM14), used only as a reference point.

The spread process uses $a = 0.5$, $\bar{s}$ equal to the sample mean (275bp), $\eta = 100$bp and $\rho = -0.3$; the capital premium is $\beta = 1\%$ per point of CET1 and the share stress is $\xi = 0.2$. Resolution recoveries follow the Markit conventions, $R = 20\%$ for Tier 2 and 40% for senior debt; the AT1 recovers nothing at PONV. The Tier 2 2035NC34 is the anchor for λ_{res}: its call (November 2034) falls inside the AT1 call window (2033–2036). Raw Capital IQ exports are kept out of the public repository; the code rebuilds a single daily panel from them.

5.4 Implied hazard and cross-sectional identification

For each date and bond j we solve $V_0^{(j)}(\lambda_j) = P_j$. The three bonds share issuer, capital and trigger, so a correct model should give $\lambda_1 = \lambda_2 = \lambda_3$. The dispersion $D = \max_j \lambda_j - \min_j \lambda_j$ is a specification test, and it can be used to ask which values of the other parameters (the spread

volatility η, the capital mean reversion κ) the three prices are consistent with. Its power is limited by the data: one basis point of hazard is worth about 0.08 points of price, so a dispersion of a few basis points is of the order of the bid–offer of evaluated prices (§7).

6 Analytics

6.1 Spread decomposition

Switch the features on one at a time and convert each price into a yield to the first call date, $y^{(k)}$ (semi-annual, bullet to T_1):

k	price $V^{(k)}$
0	certain call at T_1, full coupons, no conversion, $\lambda = 0$ (risk-free)
1	+ resolution hazard λ_{res} priced in Tier 2
2	+ AT1-specific hazard: total $\lambda = \lambda_{\text{res}} + \lambda_{\text{spec}}$
3	+ MDA coupon restriction
4	+ conversion at the trigger
5	+ economic call rule (5) = market price

Because $V^{(0)}$ is discounted at r, $y^{(0)}$ is the Treasury yield, and the increments $y^{(k)} - y^{(k-1)}$ telescope to the market yield-to-call spread. The order is chosen so that every feature is valued with risky discounting and extension comes last, with all other features in place.

6.2 Call probability and the cost of the call option

$\Pr(\tau_c = T_1)$ is read from the paths. The cost of the issuer's option is $V^{\text{call at } T_1} - V$. Negative convexity is measured as the extra loss for a +50bp shift in the spread level (with the hazard scaled accordingly), relative to a bond whose call at T_1 is certain.

6.3 Stress and capital delta

Each scenario shifts C_0, the spread (S_0 and $\bar{s}$, with λ scaled by S) or the Treasury level, and reprices on the same paths. The capital delta $\partial V / \partial C_0$ is a finite difference on a grid of C_0.

7 Results: BNP Paribas, 25 September 2026

All results are as of 25 September 2026. In the following month BNP's spreads widened with French sovereign stress (the 10-year OAT above 5%): senior CDS 5y from 57 to 74bp and, in Capital IQ spreads to benchmark, the SNP from 98 to 122bp and the Tier 2 2035NC34 from 141 to 153bp by 27 October. The analysis does not include that move.

7.1 Capital

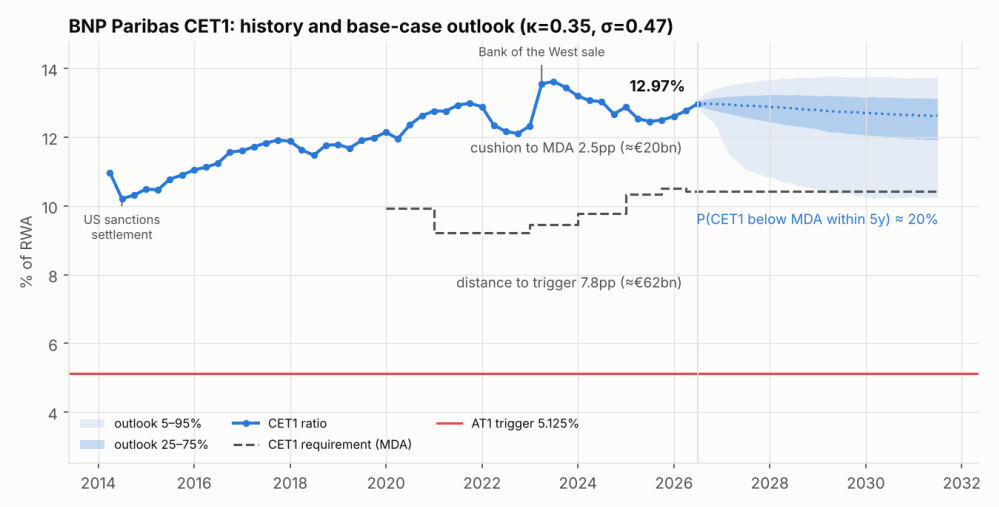


Figure 1: CET1 ratio since 2014, CET1 requirement (MDA threshold), AT1 trigger and the base-case 5-year outlook (median, 50% and 90% bands of the simulated CET1).

CET1 was 12.97% at June 2026 against a 10.43% requirement: a cushion of 2.5pp ($\approx$ EUR 20bn), which the O-SII step-ups shrink to 2.0pp by 2028 at constant capital. The trigger is 7.8pp lower ($\approx$ EUR 62bn of losses). Under the base-case dynamics, about 20% of the simulated paths fall below the MDA threshold at some point in the next five years, while the trigger is almost never reached by diffusion alone: capital risk for these bonds is first a coupon risk, then a conversion risk.

7.2 The three bonds and the implied hazard

	6.875% NC33	7.20% NC36	7.45% NC35
Reset margin (bp)	285	294	313
Clean price	94.39	95.25	97.57
Model price with $\lambda = 0$	114.2	116.9	117.6
Implied hazard λ (%/yr)	2.21	2.18	2.19
Pr(called at T_1)	49%	52%	58%
Lifetime Pr(conversion)	1.4%	1.6%	1.4%
Bid–offer (points)	0.26	0.29	0.15

Table 1: Last date, base case ($\kappa = 0.35$, $\eta = 100$bp).

Dispersion of the implied hazards across the three bonds, by spread volatility:

η (bp)	0	25	50	75	100	150	200
$D(\eta)$ (bp of hazard)	8.2	5.8	4.1	3.0	**2.6**	2.0	1.4

A stochastic refinancing spread helps (8.2bp with a deterministic one), but the dispersion keeps falling as η rises, so the cross-section does not pin η down; 100bp is kept as an assumption. Over

66 weekly dates the mean dispersion is 5.9bp against 9.3bp for the deterministic rule. With Capital IQ benchmark proxies instead of the FRED curve the last-date dispersion is 2.3–7.8bp.

These differences should not be over-read. One basis point of hazard is worth 0.08 points of price, so the last-date dispersion of 2.6bp is 0.20–0.22 points: of the size of the bid–offer of the evaluated prices (0.15–0.29). The three bonds are consistent with one hazard *within the bid–offer*; they cannot tell apart model variants whose dispersions differ by a few basis points.

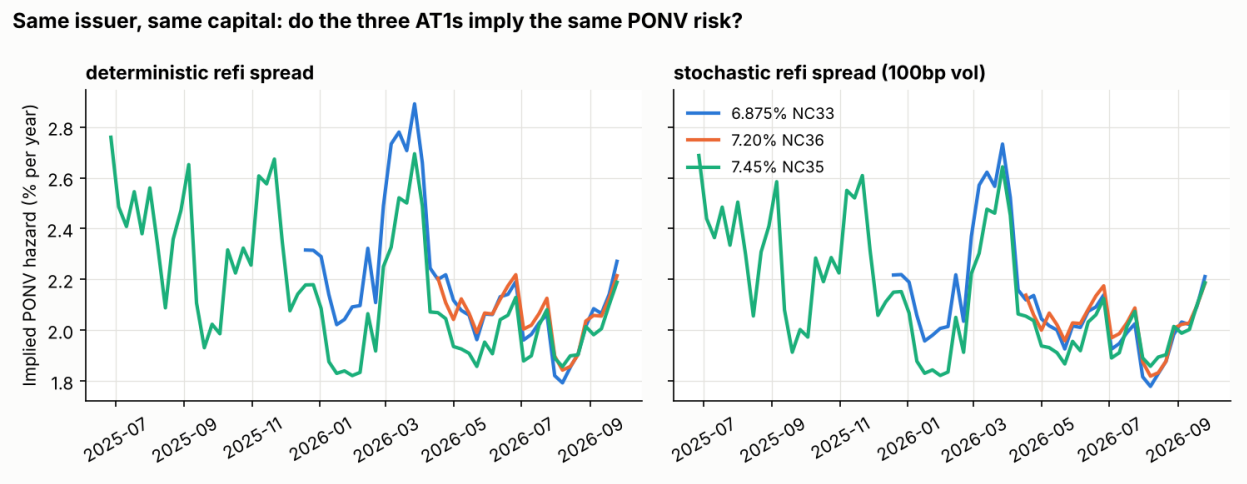


Figure 2: Implied PONV hazard per bond, weekly.

7.3 What BNP's Tier 2 says: risk priced in Tier 2 vs AT1-specific risk

	Spread (bp)	Recovery	Hazard (%/yr)	Share of AT1 hazard
Senior CDS 5Y (EUR)	57	40%	0.95	43%
Senior CDS 7Y (EUR, 27 Sep)	68	40%	1.13	52%
SNP 5.738% 2035NC34	103	40%	1.63	74%
T2 2.588% 2035NC30	116	20%	1.40	64%
T2 5.906% 2035NC34	**144**	**20%**	**1.72**	**78%**
AT1 (average of three)	274	0%	2.19	100%

Table 2: BNP's capital structure on 25 September 2026. Bond spreads are yield-to-call over the call-matched FRED Treasury; bond hazards are implied on the same simulated paths as the AT1s; CDS hazards use the credit triangle $s/(1-R)$.

The ladder is orderly: each step down the capital structure pays more, and the two instruments that share the resolution event, Tier 2 and SNP, imply nearly the same hazard (1.72% and 1.63%) under the Markit recoveries. With the SNP at 40%, the Tier 2 recovery that equalises the two is 16%, close to the conventional 20%. The CDS sits lower, partly because it is quoted in EUR at shorter tenors and is more liquid than the cash bonds.

The split. With the Tier 2 2035NC34 as anchor, in the base case,

$$\lambda = 2.19\% = \underbrace{1.72\%}_{\lambda_{\text{res}}:\ \text{priced in Tier 2}} + \underbrace{0.47\%}_{\lambda_{\text{spec}}:\ \text{AT1-specific}},$$

so about four-fifths of the AT1 hazard is compensation that BNP's Tier 2 also earns. It is mostly the risk of resolution, which the AT1 holder would also bear in Tier 2 at a better recovery; it also contains whatever risk and liquidity premia Tier 2 itself carries, so "priced in Tier 2" is the accurate label, and "bank credit risk" an approximation. Over the 66 weekly dates the Tier 2 share ranges from 66% to 80% (mean 72%). The two halves are not equally robust: λ_{res} ranges only from 1.61% to 1.74% across the bootstrap interval of the capital mean reversion, while λ_{spec} absorbs almost all of that uncertainty (§7.8). The split also depends on the recovery assumed for Tier 2:

Tier 2 recovery	0%	10%	20%	30%	40%
λ_{res} (%/yr)	1.37	1.53	1.72	1.97	2.30
λ_{spec} (%/yr)	0.82	0.67	0.47	0.23	−0.11
Share of AT1 hazard	63%	70%	78%	90%	105%

A Tier 2 recovery as high as the senior one would leave nothing for the AT1 premium, which is implausible; any recovery up to 30% attributes at least three-fifths of the AT1 hazard to what Tier 2 also prices.

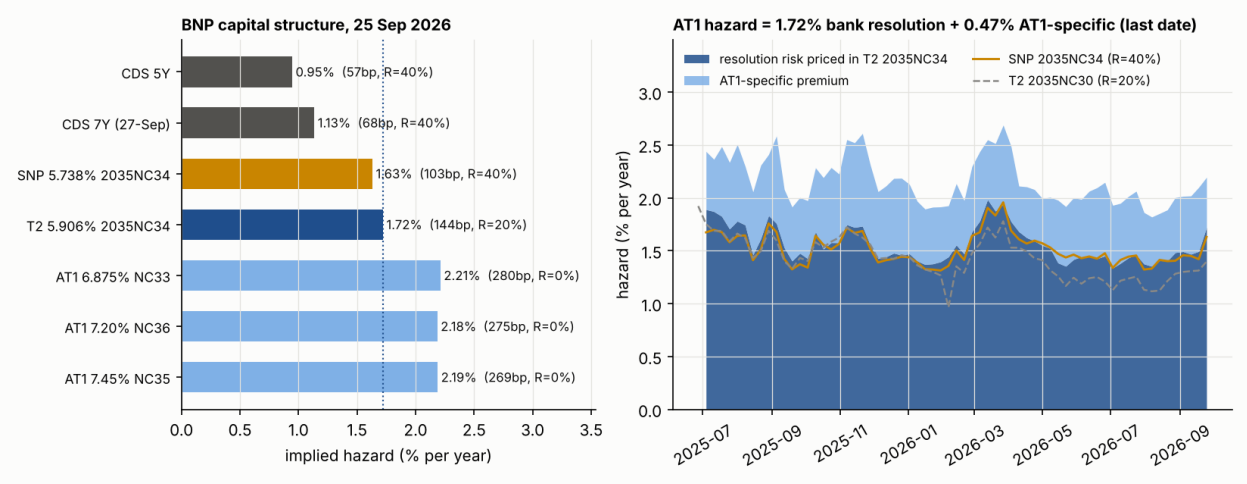
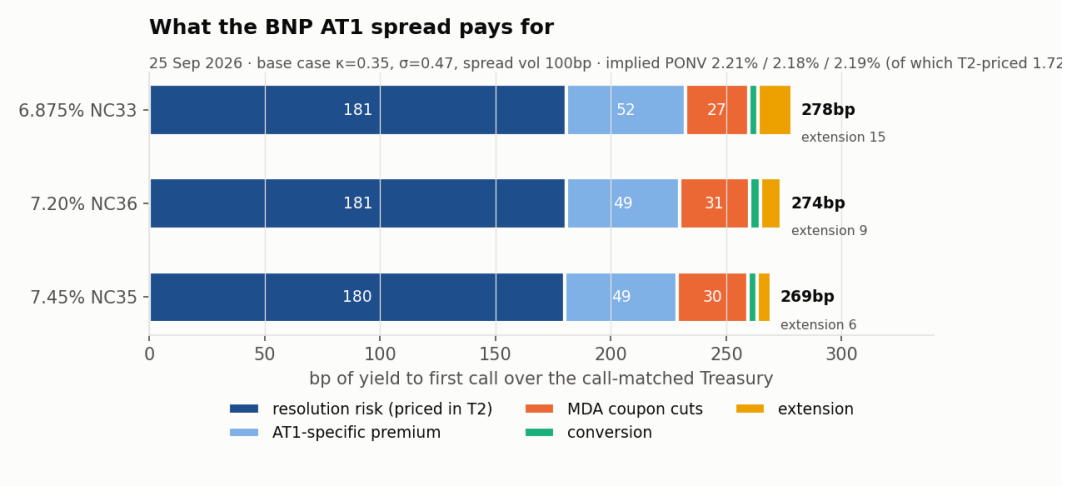


Figure 3: Left: implied hazards down BNP's capital structure. Right: the AT1 hazard (average of the three bonds), weekly, split into the part priced in the Tier 2 2035NC34 and the AT1-specific premium; the SNP and the short Tier 2 are shown for comparison.

7.4 What the spread pays for

bp of yield-to-call spread	6.875% NC33	7.20% NC36	7.45% NC35
Resolution risk (priced in Tier 2)	181	181	180
AT1-specific premium	52	49	49
MDA coupon cuts	27	31	30
Conversion at the trigger	4	4	4
Extension	15	9	6
Total = market	279	274	269

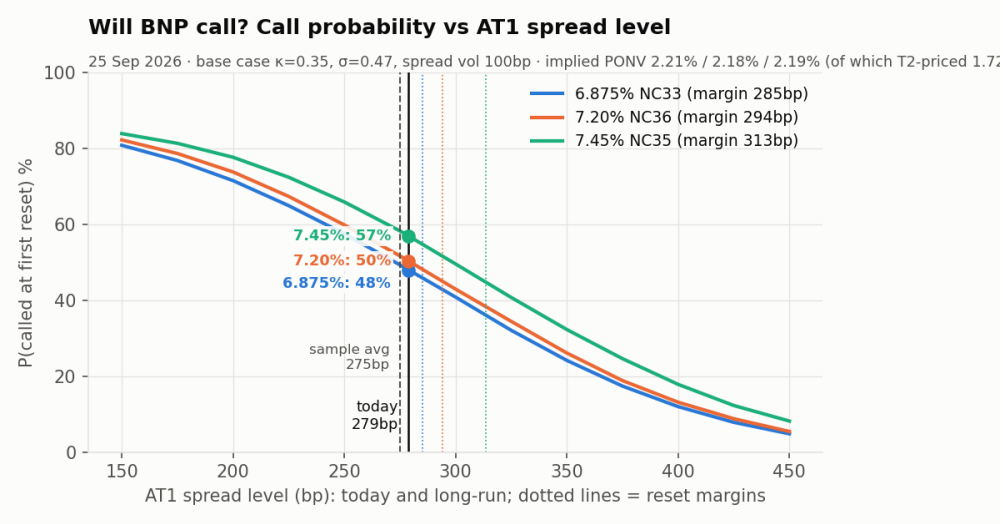


Two-thirds of the spread, about 180bp, is risk that BNP's Tier 2 also prices; in the base case another 50bp is the premium for being an AT1 rather than a Tier 2 (write-down without resolution, the equity-like perpetual and discretionary profile, liquidity). The contract features explain the rest: expected MDA coupon cuts ≈ 30bp, conversion at the trigger, a 1.4–1.6% lifetime event, ≈ 4bp, and extension 6–15bp, decreasing in the reset margin as the theory requires.

The MDA figure is an *upper bound*. The model applies the MDA quartile factor to the coupon itself, while the MDA caps all distributions and a bank inside the buffer would cut dividends, buy-backs and variable pay first. The total spread is fixed by the market, so any overstatement of the MDA component is taken from the AT1-specific premium: given κ, the 50bp is a *lower bound*. Across values of κ it is not: slower capital rebuilding raises the MDA and extension components and lowers the premium, which is close to zero at the historical estimate (§7.8). The resolution component and the 89–98bp that each AT1 pays over it do not depend on κ.

7.5 Call and extension

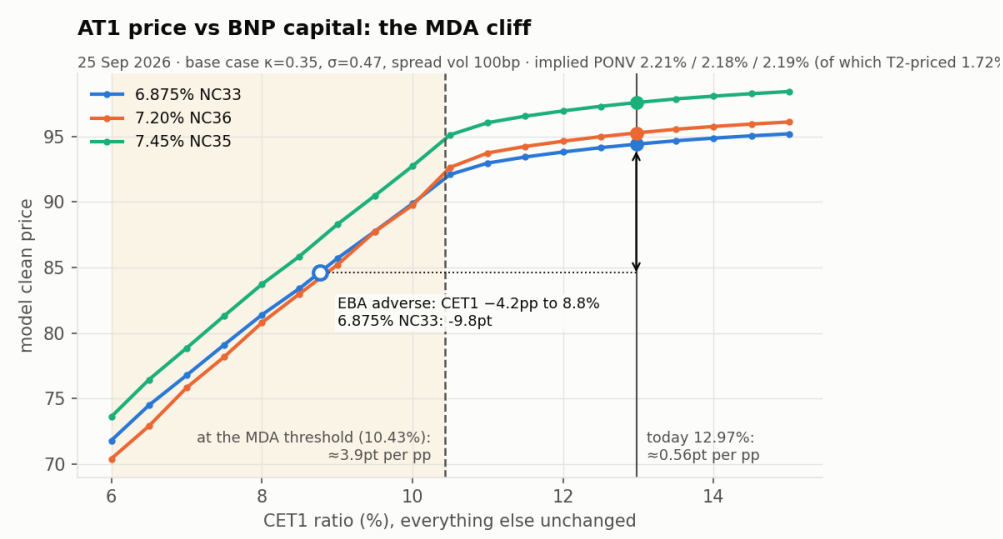
	6.875% NC33	7.20% NC36	7.45% NC35
Margin − today's AT1 spread (bp)	+6	+15	+34
Cost of the issuer's option (pts)	0.78	0.59	0.37
P&L, spreads +50bp (pts)	−3.3	−3.5	−3.2
of which from extension (pts)	−1.2	−0.9	−0.7



The low-margin bond is the most negatively convex. In the four weeks to 25 September 2026 the 10y Treasury rose 45bp; on top of that, the spread of the 6.875% widened by 34bp, against 10–12bp for the other two.

7.6 Stress and the capital cliff

Clean price change (points)	6.875% NC33	7.20% NC36	7.45% NC35
Rates +100bp	−5.3	−6.4	−6.1
CET1 −2pp	−1.5	−1.6	−1.6
CET1 −4.2pp (EBA adverse)	−9.8	−11.1	−10.4
AT1 spreads +200bp	−16.4	−16.2	−15.4
EBA adverse + spreads +200bp	−25.4	−26.6	−25.1
Bank credit: Tier 2-priced hazard ×2	−11.9	−13.0	−12.3
AT1-specific premium ×2	−3.6	−3.8	−3.6



The two hazard shocks replace an arbitrary "hazard ×3" scenario with ones that have a market counterpart: doubling the Tier 2-priced hazard is roughly what a doubling of BNP's Tier 2 spread would imply, and it costs three times as much as doubling the AT1-specific premium. The capital delta is ≈ 0.6 points per point of CET1 at 13% (CET1 ±0.5pp) and ≈ 4 at the MDA threshold (±0.25pp), the same definitions as in §7.8. Crossing $\bar{c}_{\text{MDA}}$ switches on three mechanisms at once: coupons are cut by (4), the call is blocked by (5), and the hazard rises through (3)–(7). For risk management, the distance that matters is the distance to the MDA threshold, not to the trigger.

7.7 Robustness: call behaviour, attribution order, hazard term structure

Reputational calls. The call rule (5) is purely economic, while European national champions have almost always called at the first date. Let the issuer call also when its margin is up to δ *below* the new-issue spread, $m > s_{T_k} - \delta$, and recalibrate:

Tolerance δ (bp)	0	25	50	100
Pr(called at T_1), 6.875% / 7.45%	49% / 58%	57% / 66%	65% / 72%	77% / 81%
Implied hazard, average (%/yr)	2.19	2.20	2.20	2.22
Extension component, 6.875% (bp)	15	14	13	11
Tier 2 share of the AT1 hazard	78%	78%	78%	78%

A reputational rule raises the call probabilities to the level practitioners quote, and lowers the extension component of the low-margin bond by a few basis points, but leaves the hazard and its split untouched: the market price pins the hazard down whatever the call rule.

Order of attribution. A sequential decomposition depends on the order in which features are switched on. Three alternatives, average of the three bonds (bp):

	Base order	Nested Shapley	Full Shapley	Reverse order
Priced in Tier 2	181	181	211	239
AT1-specific premium	50	50	58	68
MDA coupon cuts	29	36	35	41
Conversion	4	5	6	11
Extension	10	2	−36	−85
Total	274	274	274	274

The base order switches the hazard on first. The nested Shapley value is symmetric within the hazard block (Tier 2-priced vs AT1-specific) and within the contract features (MDA, conversion, extension), with the hazard block first. The full Shapley value and the reverse order also average in orders that value extension on a *riskless* bond, where not calling a bond that pays the reset margin over Treasuries is a gift to the investor; this is the inconsistency §3.5 and (7) are designed to avoid, and it makes the extension component negative. In every attribution in which the hazard is priced before the contract features, the Tier 2 and AT1-specific components are the same (181 and 50bp); only the split of the remaining ≈ 43bp among MDA, conversion and extension depends on the order.

Term structure of the resolution hazard. The two Tier 2 bonds call in August 2030 and November 2034. Bootstrapping a two-piece hazard instead of one flat level gives 1.40% a year to 2030 and 2.10% after (flat: 1.72%). The AT1-specific hazard re-solved on top of it falls from 0.47% to 0.39% on average, and the Tier 2-priced component of the spread moves by less than

5bp (176–185bp). An upward-sloping resolution curve slightly *raises* the share of the AT1 hazard attributed to Tier 2.

7.8 How much can we trust the capital parameters?

Bootstrap. We simulate 1,000 CET1 histories on the observed quarterly dates with the fitted dynamics and re-estimate (κ, σ) on each. The 90% intervals are $\kappa \in [0.06, 0.48]$ and $\sigma \in [0.37, 0.52]$: volatility is well estimated, the speed of capital rebuilding is not. The half-life of a capital shock, $\ln 2/\kappa$, lies between 1.5 and 11 years (point estimate 4.2).

Average of the three bonds	$\kappa = 0.06$ (5%)	$\kappa = 0.165$ (history)	$\kappa = 0.35$ (base)	$\kappa = 0.48$ (95%)
Implied PONV (%/yr)	0.66	1.64	2.20	2.33
priced in Tier 2, λ_{res} (%/yr)	1.61	1.67	1.72	1.74
AT1-specific, λ_{spec} (%/yr)	−0.95	−0.03	0.47	0.59
MDA coupon cost (pts)	14.5	6.3	2.5	1.7
Capital delta at the MDA threshold (pts/pp)	13.7	8.8	4.2	2.8
EBA adverse shock (pts)	−37.7	−21.4	−10.5	−7.3
Cross-bond dispersion D (bp)	7.8	4.2	2.6	2.3
History log-likelihood (relative to MLE)	−0.5	0	−1.5	−4.1

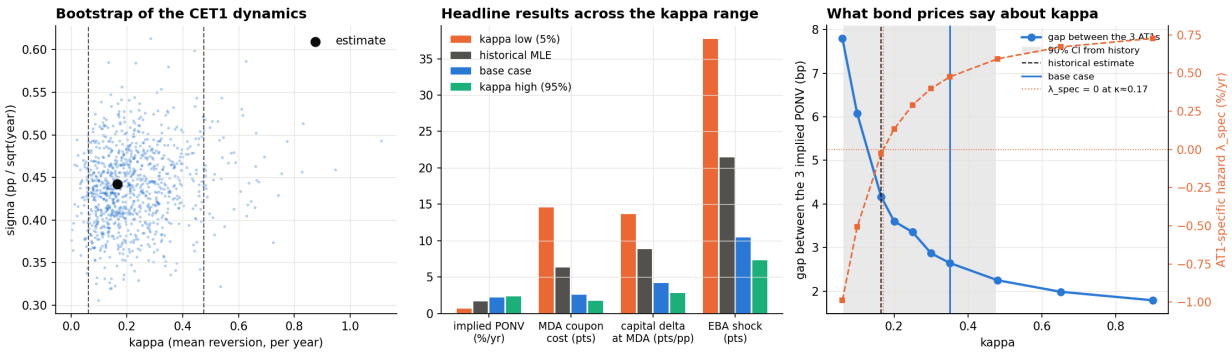


Figure 4: Left: bootstrap of (κ, σ). Centre: headline results across the κ range. Right: cross-bond dispersion of the implied hazard against κ (left axis) and the AT1-specific hazard λ_{spec} (right axis).

What prices add. History and prices pull in opposite directions. The three bonds are more consistent with fast capital rebuilding: D falls from 7.8bp at $\kappa = 0.06$ to 2.3bp at 0.48 and then flattens. In price terms that is 0.6 against 0.2 points, around the bid–offer, so the evidence is a lean, not an identification. The history, for its part, disfavours values much above 0.35 (a likelihood-ratio statistic of 3.1 at $\kappa = 0.35$, 8.1 at 0.48, crossing the 5% critical value at about 0.37).

What the Tier 2 adds. Re-solved on the paths of each κ, the hazard priced in Tier 2 barely moves, so the uncertainty about capital dynamics falls almost entirely on λ_{spec}, which is negative below $\kappa \approx 0.17$: with slow capital rebuilding the model attributes so much of the AT1's discount to expected coupon cuts that the AT1 would be priced as less risky than BNP's own Tier 2, which the creditor hierarchy rules out. Requiring $\lambda_{\text{spec}} \geq 0$ places the bound at $\kappa = 0.171$, indistinguishable

from the historical estimate, and excludes the lower half of the bootstrap interval. History and the Tier 2 together admit $\kappa \in [0.17, 0.37]$ (half-life 1.9 to 4.1 years). The base case $\kappa = 0.35$ lies in the upper part of that range, where the cross-section of AT1s also points, and is consistent with BNP's active capital management (a 13% target, managed RWA and distributions). Across the range (average of the three bonds, bp):

	$\kappa = 0.171$	$\kappa = 0.35$ (base)	$\kappa = 0.37$
Priced in Tier 2	180.1	180.5	180.5
AT1-specific premium	1.0	49.8	52.3
MDA coupon cuts	52.0	29.4	27.9
Conversion	6.7	4.3	4.1
Extension	34.0	10.0	9.1
AT1 over Tier 2	93.8	93.4	93.4

The Tier 2-priced component and the total the AT1 pays over it move by less than half a basis point. Their split does not: at $\kappa = 0.17$ the premium vanishes, and expected coupon cuts and extension take its place, because CET1 is more often below the MDA threshold on a reset date and the call is blocked more often. Two caveats apply. The bound is conditional on the stylised MDA rule, which overstates the coupon loss most when capital is slow to rebuild, so the true bound lies below 0.17. And $\lambda_{\text{spec}} = 0$ is the weakest restriction the hierarchy imposes; any positive liquidity or equity premium would raise it.

The ordering of the bonds, the extension ranking and the steepening of the capital delta near the MDA threshold hold across the whole range. The level of the capital stress and the split of the AT1's excess over the Tier 2 do not, and are reported with this range.

Where the uncertainty sits. Each curve in Figure 5 is re-calibrated to today's price of the 6.875%, so all four scenarios agree at today's 13% CET1 (94.4). They separate only as capital falls: at the MDA threshold the price ranges from 76 ($\kappa = 0.06$) to 93 ($\kappa = 0.48$), and at 6% CET1 from 36 to 77. Prices identify the level of risk today, not the shape of the cliff.

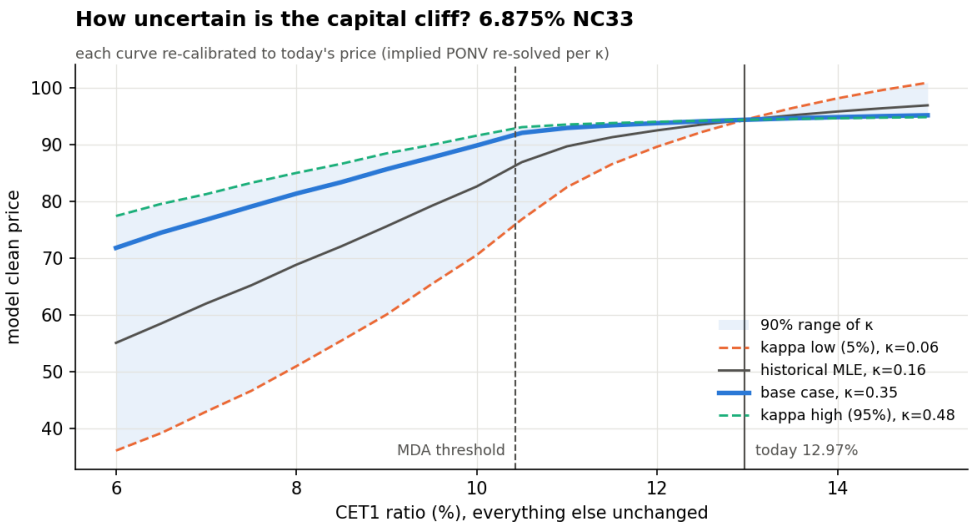


Figure 5: Price of the 6.875% NC33 against CET1 for four values of κ, each re-calibrated to today's price. Shaded: 90% bootstrap range of κ.

8 Limitations and extensions

1. **Measure.** CET1 is not traded. Its dynamics are estimated under the physical measure, and all remaining risk premia are absorbed by λ, which is therefore a risk-neutral hazard, not a default probability. The Tier 2 anchor splits it, but λ_{res} is risk-neutral too.

2. **Recovery at resolution.** λ_{res} depends on the Tier 2 recovery (§7.3); with the senior bond at 40%, the market data are consistent with a Tier 2 recovery near the conventional 20%, but neither is observed.

3. **Capital mean reversion.** κ is weakly identified by 50 quarters of history and drives the capital stress results and the split of the AT1's excess over the Tier 2 (§7.8). The Tier 2 restriction $\lambda_{\text{spec}} \geq 0$ excludes the slower half of the range but is itself conditional on the stylised MDA rule; the Tier 2-priced component does not depend on κ.

4. **MDA mechanics.** The MDA caps *all* distributions (dividends, buy-backs, variable pay, AT1 coupons) at a fraction of distributable profit; a bank inside the buffer would cut dividends first. Applying the quartile factor to the coupon itself, as (4) does, overstates the MDA component. The binding constraint may also be the leverage-ratio or MREL MDA, and available distributable items, none of which is modelled.

5. **Call behaviour.** The base call rule is purely economic; a reputational tolerance raises the call probabilities but not the hazard or its split (§7.7). The approval test under CRR art. 77–78 is not the MDA test.

6. **Systemic and sovereign risk.** The spread process is issuer-specific and capital jumps do not move spreads. AT1 spreads widened by hundreds of basis points in March 2020 and March 2023 with no change in capital, and after the valuation date BNP's senior spreads widened with French sovereign stress (CDS 5y from 57 to 74bp by 27 October 2026). A common sector/sovereign factor with co-jumps is the main extension.

7. **Rates.** Today's FRED curve is used for all future resets; a short-rate model (Hull–White) would add rate-driven extension.

8. **Spread volatility.** η is not pinned down by the cross-section.

9. **Capital premium.** β controls the size of the capital cliff and is assumed. A cross-bank regression of AT1 spreads on MDA distance would calibrate it.

10. **Data.** Prices are evaluated rather than traded, cover 15 months of tight spreads, and the bid–offer is of the size of the cross-bond hazard dispersion. TRACE prints would test the results on transactions. The CDS is quoted in EUR, the bonds in USD.

11. **Conversion.** A single stressed share price replaces a joint equity–capital process, and PONV recovery is set to zero.

A Parameters

Symbol	Meaning	Value	Source
κ, σ	CET1 mean reversion, volatility	0.35, 0.47pp	MLE 0.165/0.44; base (§7.8)
θ	CET1 target	13%	BNP strategic plan
λ_J, m_J, σ_J	stress jumps	0.12, 2.2pp, 0.5	EBA 2025 check
b, w	MDA floor, combined buffer	5.64%, 4.79% (+0.25pp 2027, 2028)	requirement, O-SII schedule
c_{trig}	trigger	5.125%	term sheets
$a, \bar{s}, \eta, \rho$	spread OU	0.5, 275bp, 100bp, −0.3	sample, assumption
β	capital premium	1% per pp	assumption
ξ	share at trigger / today	0.2	assumption
λ	PONV hazard today	$\approx 2.2\%$	implied (AT1s)
λ_{res}	resolution hazard	1.7%	implied (T2 2035NC34)
$R_{\text{T2}}, R_{\text{SNP}}$	recovery at resolution	20%, 40%	Markit conventions

B Code map

File	Content
`processes.py`	CET1 jump-diffusion (1), trigger times
`instrument.py`	MDA schedule (4), stylised term sheet
`pricer.py`, `risk.py`	original engine, Greeks, loss distribution
`market.py`	dated bonds, (2)–(7), pricing (11), implied λ; Tier 2/SNP bullets (9), implied λ_{res}
`data.py`, `build_bnp_dataset.py`	data layout, Capital IQ exports to one panel
`calibrate_bnp.py`	calibration (§5)
`capital_structure.py`	Tier 2/SNP/CDS anchor (§7.3)
`robustness.py`	call tolerance, attribution order, hazard term structure (§7.7)
`spread_and_stress.py`	decomposition, call analysis, stress
`parameter_uncertainty.py`	bootstrap, κ sensitivity (§7.8)